

Benyamin Tabarsi

Computer Scientist / AI and CS Education Researcher

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Research Interests

My research connects artificial intelligence and computing education, with an emphasis on designing and evaluating intelligent tools that support learners and educators. I am particularly interested in human-AI co-intelligence, understanding how students, developers, and educators collaborate with AI systems, and how we can design those systems to be trustworthy, adaptive, and pedagogically grounded. My work also draws on educational data mining to identify novice programmers' struggles and develop learning supports. In addition, I investigate how AI can be extended into health communication as a sensitive domain, where responsible AI design is especially critical. My long-term goal is to leverage AI for social impact by developing practical technologies that make educational resources more accessible and effective.

Education

North Carolina State University Raleigh, NC	PhD in Computer Science Expected May 2027 <i>Advisors: Dr. Tiffany Barnes, Dr. Dongkuan (DK) Xu</i>
Science and Research Branch Azad University Tehran, Iran	Master's in Computer Software Engineering 2019
University of Mazandaran Babolsar, Iran	Bachelor's in Information Technology Engineering 2016

Research Experience

Graduate Research Assistant Game2Learn and GIC Labs North Carolina State University May 2022 – Present	• Conducting multiple studies on Generative AI, including design and research on a course-specific AI assistant, exploring developers' usage of LLMs, co-development of a RAG-based healthcare app, and training educators on integrating ChatGPT into classrooms • Performed studies on coding patterns of novice programmers to boost their learning experience and develop intelligent support tools
Research Assistant Distributed Systems Lab Azad University Nov 2017 – Aug 2019	• Studied latency-power trade-off and resource management in fog-based IoT networks • Guided students in conducting research and led group discussions on topics concerning distributed systems, IoT, and e-healthcare

Publications

Conference and Workshop Papers

- [8] Heidi Reichert, **Benyamin Tabarsi**, Zifan Zang, Cheri Fennell, Indira Bhandari, David Robinson, Madeline Drayton, Catherine Crofton, Matthew Lococo, Dongkuan (DK) Xu, Tiffany Barnes (2024). "Empowering Secondary School Teachers: Creating, Executing, and Evaluating a Transformative Professional Development Course on ChatGPT". In *Proceedings of the 2024 IEEE Frontiers in Education (FIE 24)*.
- [7] Heidi Reichert, **Benyamin Tabarsi**, Thomas Price, Tiffany Barnes (2024). "Jigsaw: A Visual Tool for Decomposing and Planning Programming Problems". In *Proceedings of the 2024 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. **Best Research Paper Award** 🏆.
- [6] **Benyamin Tabarsi**, Heidi Reichert, Nicholas Lytle, Veronica Catete, Tiffany Barnes (2024). "Scaffolding Novices: Analyzing When and How Parsons Problems Impact Novice Programming in an Integrated Science Assignment". In *Proceedings of the 2024 ACM Conference on International Computing Education Research (ICER)*.
- [5] Wengran Wang, John Bacher, Amy Isvik, Ally Limke, Sandeep Sthapit, Yang Shi, **Benyamin T. Tabarsi**, Keith Tran, Veronica Cateté, Tiffany Barnes, Chris Martens, Thomas Price (2023). "Investigating the Impact of On-Demand Code Examples on Novices' Open-Ended Programming Experience". In *Proceedings of the 2023 ACM Conference on International Computing Education Research (ICER)*.
- [4] **Benyamin Tabarsi**, Heidi Reichert, Rachel Qualls, Thomas Price, Tiffany Barnes (2023). "Exploring Novices' Struggle and Progress During Programming Through Data-Driven Detectors and Think-Aloud Protocols". In *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*.
- [3] Wengran Wang, Gordon Fraser, Mahesh Bobbadi, **Benyamin T. Tabarsi**, Tiffany Barnes, Chris Martens, Shuyin Jiao, Thomas Price (2022). "Pinpoint: A record, replay, and extract system to support code comprehension and reuse". In *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*.
- [2] **Benyamin T. Tabarsi**, Ally Limke, Heidi Reichert, Rachel Qualls, Thomas Price, Chris Martens, Tiffany Barnes (2022). "How to Catch Novice Programmers' Struggles: Detecting Moments of Struggle in Open-Ended Block-Based Programming Projects using Trace Log Data". In *Proceedings of the 6th Educational Data Mining in Computer Science Education (CSEDM) Workshop*.
- [1] Heidi Reichert, Ally Limke, **Benyamin T. Tabarsi**, Thomas Price, Chris Martens, Tiffany Barnes (2022). "How, when, and why do novices struggle in programming? Exploring the experiences and perceptions of common programming moments in block-based environments". In *Proceedings of the 6th Educational Data Mining in Computer Science Education (CSEDM) Workshop*.

Journal Papers

- [3] **Benyamin Tabarsi**, Heidi Reichert, Sam Gilson, Ally Limke, Sandeep Kuttal, Tiffany Barnes (2026), "LLMs' Reshaping of People, Processes, Products, and Society in Software Development: A Comprehensive Exploration with Early Adopters". *Under Revision*.
- [2] **Benyamin Tabarsi**, Tahreem Yasir, Heidi Reichert, Xiaoyi Tian, Shiva Gadireddy, Clara DiMarco, Daniel Briceno, Tiffany Barnes (2025). "Herald of Advancement, Disruption, or Both: A Systematic Literature Review on Student-Facing LLM Tools in Undergraduate Computing Education". *Under Review*.
- [1] **Benyamin T. Tabarsi**, Ali Rezaee, Ali Movaghar (2022). "ROGI: Partial Computation Offloading and Resource Allocation in the Fog-Based IoT Network Towards Optimizing Latency and Power Consumption". In *Cluster Computing*.

Posters and Demos

- [4] **Benyamin Tabarsi** (2026). "MerryQuery in the Loop: How Students Use and Perceive a GenAI-Powered Assistant in an Introductory Programming Course." In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V. 2 (SIGCSE)*.
- [3] **Benyamin Tabarsi** (2025). "Developing LLM-Powered Trustworthy Agents for Personalized Learning Support." In *Proceedings of the 39th Association for the Advancement of Artificial Intelligence (AAAI)*.

- [2] **Benyamin Tabarsi**, Aditya Basarkar, Xukun Liu, Dongkuan (DK) Xu, Tiffany Barnes (2025). "MerryQuery: A Trustworthy LLM-Powered Tool Providing Personalized Support". In *Proceedings of the 39th Association for the Advancement of Artificial Intelligence (AAAI)*. **Best Technical Demonstration Award** 🏆.
- [1] Heidi Reichert, Sandeep Sthapit, **Benyamin T. Tabarsi**, Ally Limke, Thomas Price, Tiffany Barnes (2024). "Experience Helps, but It Isn't Everything: Exploring Causes of Affective State in Novice Programmers". In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2 (SIGCSE)*.

Honors and Awards

- **Preparing the Professoriate (PTP) Fellow, NC State University** 2025 – 2026
Competitive, one-year faculty-mentored program focused on faculty mentorship, pedagogy seminars, and co-teaching experience. Mentored by Dr. Lina Battestilli for CSC 113 (Intro to Programming - MATLAB)
- **Best Technical Demonstration Award in AAAI** 2025
For "MerryQuery: A Trustworthy LLM-Powered Tool Providing Personalized Support for Educators and Students"
- **Best Research Paper Award in IEEE VL/HCC** 2024
For "Jigsaw: A Visual Tool for Decomposing and Planning Programming Problems"
- **Ranked #1 in Cumulative GPA**
Highest cumulative GPA in the 2016 cohort of the Master's program in Computer Software Engineering.

Grants and Funding

- **SIGCSE Travel Grant, NC State University** 2026
Travel Grant for participation in SIGCSE 2026
- **Mentored Teaching Fellow, NC State University** Spring 2026
Awarded \$1,000 for co-instruction and application of teaching strategies under faculty mentorship (Dr. Lina Battestilli)
- **DELTA Grant, NC State University** 2025 – 2026
Collaborated on an \$8,000 grant proposal (with Dr. Tiffany Barnes, Dr. DK Xu, and Dr. Lina Battestilli) to enhance and continue the development of MerryQuery.
- **Researcher under NSF CCRI, NSF Creative CS, and RET Projects, NC State University**
Continuous Research Assistantship under NSF projects in the Game2Learn lab for 3.5 years
- **AAAI 2025 Student Travel Grant (\$500)**
Travel Grant for participation in AAAI 2025 for papers in doctoral consortium and demo tracks
- **Computer Science Department Travel Fund (\$460), NC State University**
Supplementary Travel Fund for participation in AAAI 2025
- **SPLICE Grant**
Grant for participation in the Educational Data Mining (2023) and Computing Education Research (2024) tracks at Carnegie Mellon University's LearnLab Summer School

Teaching Experience

- **How to Integrate LLMs into K-12 Classrooms** (Instructor and Mentor), NC State University Jun 2023 – Jun 2024
Mentored the development and co-led instruction of a workshop series for K-12 teachers on integrating ChatGPT into a classroom across three iterations

- **Introduction to Computing Environments** (Instructor), *NC State University* May – Aug 2023
Administered course, coordinated curriculum implementation, student support, and supervised teaching assistants
- **Art, Coding, Action!** (Teacher and Curriculum Consultant), *NC State University* Jul 2022
Led 20+ high school students in a one-week block-based programming camp, focusing on games and art in Snap!
- **Automated Learning and Data Analysis** (Teaching Assistant), *NC State University* Jan 2022 – May 2022
Assisted in designing coding/written questions, grading, and office hours
- **Data Structures and Algorithms** (Teaching Assistant), *NC State University* Aug 2021 – Dec 2021
Graded assignments/exams and provided academic support during office hours

Invited Talks and Panels

- **Graduate Life and PhD Experience** - Panel Fall 2025
Shared insights and answered questions about graduate school, and the future of computer science and programming in the age of AI.
- **MerryQuery** - Guest Talk Fall 2025
Discussed AI in education and MerryQuery's current status in the "Natural Language Processing" course at Utah State University
- **MerryQuery** - Guest Talk Fall 2025
Presented early findings and the current status of MerryQuery at the DELTA AI Roundtable "IDIG: AI in Education" Roundtable.
- **AI in Education** - Podcast Summer 2025
Discussed Opportunities and Challenges of AI in Education and MerryQuery's development and research journey
- **MerryQuery** - Guest Lecture Fall 2024
Presented to the students of the course "Intro to Educational Innovation & Entrepreneurship" at NC State University
- **PhD in CS Experience** - Panel Spring 2024
Shared insights on PhD experience, balancing research and coursework, and strategies for success in graduate school at NC State Doctoral Recruiting Day
- **How to Make Research Questions** - Workshop Summer 2023 and 2024
Taught undergraduate research interns how to develop focused, feasible research questions for their summer projects
- **How to Present Research** - Workshop Summer 2023
Guided undergraduate research interns on effective poster and oral presentation techniques
- **From the Internet of Things to Fog Computing** - Guest Lecture 2018
Presented the evolution of distributed systems from IoT to fog computing to the "Distributed Systems" course students

Mentorship Experience

Undergraduate Research Interns

MerryBench

1. Jayani Sivakumar (*North Carolina State University, NC*) Summer 2025
2. Susanna Quayle (*Fayetteville Technical Community College, NC*) Spring 2025, Summer 2025

MerryQuery

1. Daniel Briceno Spring 2026
1. Aishwarya Radhakrishnan (*North Carolina State University, NC*) Summer 2024 and Fall 2024
2. Praneel Magapu (*North Carolina State University, NC*) Summer 2024

3. Aditya Basarkar (North Carolina State University, NC)	Summer 2024
4. Lavan Aditya (North Carolina State University, NC)	Summer 2024

SafeTalkCoach

1. Ary Kumar (North Carolina State University, NC)	Fall 2024, Spring 2025
2. Michelle Jiang (UNC Chapel Hill, NC)	Summer 2024
3. Jonathan Hardwick (Fayetteville Technical Community College, NC)	Summer 2024
4. Susanna Quayle (Fayetteville Technical Community College, NC)	Summer 2024

Systematic Literature Review on LLMs and Computing Education

1. Shiva Gadireddy (North Carolina State University, NC)	Summer 2023, Fall 2023, and Spring 2024
2. Lavan Aditya (North Carolina State University, NC)	Spring 2024
3. Samantha Gonzalez (Kean University, NJ)	Summer 2023
4. Yadhira Marcos-Avila (UNC Charlotte, NC)	Summer 2023

Detectors for Identifying Novices' Struggle during Programming

1. Rachel Qualls (University of Alabama, AL)	Summer 2022
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Participatory Design on a Block-Based Programming Learning Management System (SnapClass)

1. Maggie Lin (North Carolina State University, NC)	Summer 2022
2. Sana Mahmoud (North Carolina State University, NC)	Summer 2022

Master's Students

MerryQuery

1. Shreyas Visweshwaran (North Carolina State University, NC)	Fall 2025
2. Teddy Chen (North Carolina State University, NC)	Fall 2024
3. Homak Patel (North Carolina State University, NC)	Fall 2024

Teacher Research Interns

Integrating LLMs into K-12 Classrooms

1. Cherri Fennel (Durham Public Schools, NC)	Summer 2023
2. Matthew Lococo (Greene County School, NC)	Summer 2023

Current Research Projects

- MerryQuery: A GenAI-Powered Tool for Personalized Educational Support**
May 2024 – Present
 Conducting research and leading the development of a GenAI-powered assistant that supports educators and students through adaptive, rule-based, and reliable AI-driven guidance.
- ParenTeenTalk: Benchmark Dataset for Parent-Teen Health Conversation**
Aug 2024 - Present
 Developing a benchmark dataset of LLM-generated conversations for parental training in sexual health communication, and an evaluation framework for assessing AI performance in sensitive health communication contexts.

Professional Service and Memberships

Program Committee Member

ACM conference on Innovation and Technology in Computer Science Education (ITiCSE)	2026
ACM Conference on International Computing Education Research (ICER)	2026
Association for the Advancement of Artificial Intelligence (AAAI)	2026
ACM Special Interest Group (SIG) on Knowledge Discovery and Data Mining (KDD)	2025 - 2026
International Joint Conference on Artificial Intelligence (IJCAI)	2025
IEEE Frontiers in Education (FIE)	2024 - 2025
International Conference on Artificial Intelligence in Education (AIED)	2025

AAAI 2025 Spring Symposium: GenAI@Edge: Empowering Generative AI at the Edge	2025
ACM The International Conference on Learning Analytics & Knowledge (LAK)	2025
IEEE Global Engineering Education Conference (EDUCON)	2025
ACM Technical Symposium on Computer Science Education (SIGCSE)	2023 - 2025

Journal Reviewer

Empirical Software Engineering
IEEE Transactions on Learning Technologies

Memberships

IEEE Student Member	2018 & 2023 – Present
ACM Student Member	2023 – Present
Sigma Xi	2025 – Present

Skills

Programming: Python, Java, C++, R, HTML, CSS, PHP, MATLAB

Tools/Libraries: GIT, LlamaIndex, LangChain, Keras, PyTorch, Matplotlib, Jupyter, NLTK, Docker, Scikit-learn

Databases and Operating Systems: MySQL, Linux, MongoDB

Research (Computing Education): A/B testing, Qualitative Coding, Survey Design, Educational Data Mining

Research (Generative AI): Prompt Engineering, Retrieval-Augmented Generation, Guardrail Design, Preference Optimization, Fine-Tuning